

Comparative Analysis of Machine Learning Models and DemoIccMAX tools in CMYK - CIELAB Color Space Transformation

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Abstract - This study investigates the efficacy of four machine learning models - ANN, DNN, SVR, and RBF - in performing color space transformations between CMYK and LAB. Using the FOGRA51 dataset, the models are evaluated for both forward (CMYK $\rightarrow$ LAB) and reverse (LAB $\rightarrow$ CMYK) transformations. The performance is benchmarked against the ICC profile utilizing the DemoIccMAX tool. Metrics such as CIE1976 ΔE , CIE2000 ΔE , and smoothness are applied to assess accuracy and practical utility.

Index Terms—Color Space Transformation, CMYK, CIELAB, DemoIccMAX

I. INTRODUCTION

A. Background and Importance of Color Management

Color management is a critical process in ensuring consistent and accurate color representation across different devices, including monitors, printers, cameras, and projectors. In industries such as graphic design, printing, and photography, precise color reproduction is essential for maintaining fidelity and achieving professional - quality results. The International Color Consortium (ICC) standards have provided a framework for managing color consistency, enabling cross-device color transformation through profiles tailored to specific devices.

B. Overview of CIELAB and CMYK Color Spaces

The CIELAB and CMYK color spaces represent two distinct approaches to color representation, each tailored to specific applications:

- **CIELAB (Lab)**:** A device-independent color space designed for uniformity, based on the CIE 1931 XYZ color space. It represents colors with three coordinates: L (lightness), a^* (green-red component), and b^* (blue-yellow component). Its uniformity makes it suitable for comparing colors across devices [4].
- **CMYK:** A device-dependent color space primarily used in the printing industry, relying on subtractive color mixing of cyan (C), magenta (M), yellow (Y), and black (K). Each color component is influenced by the specific characteristics of the substrate and printing process.

C. Transformation and Limitations of Traditional Methods

Color space transformation, the process of converting colors from one representation to another, is central to color management. Traditional methods such as polynomial regression and Look-Up Tables (LUT) have been widely adopted:

- **LUTs:** These are precomputed mappings stored in a table format, enabling rapid transformations between color

spaces. While effective, LUTs are prone to interpolation errors and may struggle with complex transformations.

- **Poly Regression:** Useful for simpler transformations (e.g., RGB to XYZ), but limited in handling the complexity of CMYK to CIELAB conversions.

Recent advancements in machine learning have introduced more flexible and accurate approaches, such as Support Vector Regression (SVR), Artificial Neural Networks (ANN), and Deep Neural Networks (DNN). These methods promise better accuracy and adaptability for complex transformations [3]

D. Previous Work

Past research has explored various machine learning models for color space transformation:

- **SVR:** Demonstrated high accuracy in RGB to CIELAB transformations.
- **ANN:** Used for RGB to CIELAB transformations, outperforming LUT-based ICC profiles in terms of smoothness and precision [2].
- **DNN:** Proven effective for multi-layer transformations like RGB to XYZ and LAB [5].
- **RBF:** Efficient in regression problems, particularly for transformations requiring non-linear mapping [6].

Despite these advancements, limited work has focused on comparing machine learning models with ICC profiles for CMYK to CIELAB transformations. Moreover, challenges like the non-uniqueness of solutions in LAB to CMYK transformations, particularly for the black (K) channel, remain underexplored.

E. Objectives and Research Questions

This project aims to address the gaps in existing research by:

- Comparing four machine learning models - ANN, DNN, SVR, and RBF - with ICC profiles for color space transformations.

- Evaluating these models using the DemoIccMAX tool, a robust platform for applying ICC profiles.
- Analyzing model performance across forward (CMYK $\rightarrow$ LAB), reverse (LAB $\rightarrow$ CMYK), and round-trip (LAB $\rightarrow$ CMYK $\rightarrow$ LAB) transformations.

The central research question is: **How do machine learning models compare with ICC profiles in terms of accuracy, smoothness, and practical application for CMYK and CIELAB color space transformations?**

F. Significance of the Study

This study is crucial for advancing color management practices by integrating state-of-the-art machine learning techniques with established ICC profiles. The findings have practical implications for industries reliant on precise color reproduction, such as printing and digital media, and contribute to the growing body of knowledge in machine learning-driven color management.

II. METHODOLOGY

A. Dataset

The FOGRA51 dataset is the foundation of this experiment, serving as a standard reference for color space transformations. It is recognized for its compliance with the **ISO 12647-2:2013** printing process definition. The dataset comprises color samples represented in both **CMYK** and **LAB** color spaces. Each sample is associated with a premium coated substrate (**PS 1, 115 g/m²**) with moderate fluorescence characteristics defined by **ISO 15397 (8-14 ΔB)**.

The dataset is preprocessed to normalize the color values for computational efficiency and model compatibility. **LAB** values are scaled to their respective ranges: L^* is scaled to [0, 1] from [0, 100], while a^* and b^* are scaled from their original range of [-128, 127] to [0, 1]. Similarly, **CMYK** values are normalized to fall within [0, 1], reflecting their percentage representation in the original data. After preprocessing, the dataset is divided into three subsets: **70%** for training, **15%** for validation, and **15%** for testing, ensuring robust model evaluation.

B. Models

Four machine learning models were selected for the color space transformations due to their proven efficacy in regression tasks: Artificial Neural Network (ANN), Deep Neural Network (DNN), Support Vector Regression (SVR), and Radial Basis Function (RBF). These models were configured separately for forward (CMYK $\rightarrow$ LAB) and reverse (LAB $\rightarrow$ CMYK) transformations.

For the forward transformation, the ANN employs a single hidden layer with 180 neurons activated by ReLU, while the output layer uses a linear activation function. The DNN architecture includes three hidden layers with 21, 77, and 21 neurons, respectively, employing ReLU activation. SVR utilizes a MultiOutputRegressor with an RBF kernel and the hyperparameters optimized for color transformation tasks. The

RBF model uses Gaussian activation functions, with its centers derived from training data and a spread parameter set to 70.

The reverse transformation introduces a dual-network architecture to address the unique challenges of predicting the K (black) channel. The CMY network processes LAB inputs to predict CMY values, and its output, combined with LAB inputs, serves as input to the K network for K value prediction. This dual structure ensures improved accuracy for LAB $\rightarrow$ CMYK transformations.

C. DemoIccMAX Tools

The **DemoIccMAX** tool was utilized to benchmark the performance of the models against ICC profiles. This open-source software suite supports advanced color management applications and facilitates the practical application of **iccMAX** profiles.

The `iccApplyNamedCmm` command-line tool was employed to apply the ICC profile transformation to the **CMYK** data. The tool was configured to use a 16-bit data encoding (`icEncode16bit`) to minimize rounding errors during **CIELAB** calculations. The tetrahedral interpolation method was selected for color transformation to ensure smoothness. The absolute colorimetric rendering intent was used for accurate color representation under specific conditions. The ICC profile, `PS0coated_v3.icc`, corresponds to the **FOGRA51** printing conditions and the **ISO 12647-2:2013** substrate standards.

D. Evaluation Metrics

The models were evaluated using metrics tailored to the nature of color transformations:

- 1) **Forward Transformation (CMYK $\rightarrow$ LAB):** Accuracy was measured using two standard color difference formulas, CIE1976 ΔE and CIE2000 ΔE , which assess perceptual and precise color differences, respectively.
- 2) **Reverse Transformation (LAB $\rightarrow$ CMYK):** Model performance was evaluated using mean absolute error (MAE) for the individual CMYK components and the Euclidean distance for combined error measurement.
- 3) **Round-trip Transformation (LAB $\rightarrow$ CMYK $\rightarrow$ LAB):** This workflow included accuracy assessments via CIE1976 ΔE and CIE2000 ΔE , as well as smoothness analysis of predicted color ramps to evaluate the continuity and perceptual consistency of color transformations.

Smoothness evaluates the perceptual continuity of color transformations using Green's (2008) framework [1]. A color ramp undergoes transformation, and the **CIEDE2000** metric is calculated between consecutive triplets in the transformed ramp, producing a ramp of length $n-1$. The second derivative of this ramp, derived by subtracting consecutive **CIEDE2000** values, results in an $n-2$ ramp, which quantifies abrupt changes. Summary statistics, such as the mean and maximum of this derivative, measure smoothness, with lower values indicating better perceptual consistency. This metric highlights transitions' continuity, complementing accuracy metrics like **CIEDE2000** and **MAE** for a comprehensive transformation evaluation.

E. Forward and Backward Workflow

For the forward workflow, the models take normalized CMYK inputs and output LAB values. The process involves applying the learned regression functions or mappings to transform the color space. The output is validated against test data using the ΔE metrics to ensure accuracy.

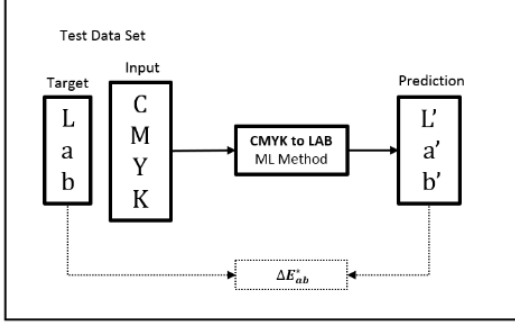


Fig. 1: Forward transformation accuracy measurement.

The backward workflow follows a dual - network design to predict CMYK values. The CMY network processes LAB values to produce CMY predictions, which are then combined with the LAB inputs for the K network. This structure addresses the complexity of black channel prediction, ensuring more precise LAB $\rightarrow$ CMYK transformations.

F. Round - trip Transformation Workflow

The round-trip workflow evaluates the combined accuracy of LAB $\rightarrow$ CMYK $\rightarrow$ LAB transformations. This workflow simulates real-world applications where an image might undergo multiple transformations during printing and display processes. The predicted LAB values from the reverse transformation are compared with the original LAB inputs to calculate ΔE metrics. Smoothness analysis is applied to assess the perceptual continuity of the transformations, particularly focusing on the color ramps generated during the forward and backward processes.

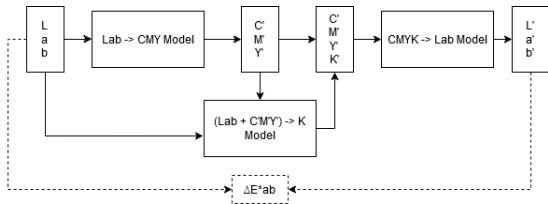


Fig. 2: Round-trip transformation accuracy measurement.

III. RESULTS

A. Forward Transformation Results Analysis

95th Percentile ΔE^* indicates the upper boundary of prediction errors while excluding outliers. Among the machine learning models, the **Radial Basis Function (RBF)** network

Metric	ANN	DNN	SVR	RBF	DemoIccMax
95th Percentile ΔE^*	2.24	3.12	2.76	1.53	0.24
Median ΔE^*	0.78	1.41	1.16	0.72	0.05
Max ΔE^*	4.21	6.7	3.95	2.38	0.51
$\Delta E^* < 3$ (%)	97.94%	93.42%	96.71%	100.00%	100.00%
$3 \leq \Delta E^* < 6$ (%)	2.06%	6.17%	3.29%	0.00%	0.00%
$\Delta E^* \geq 6$ (%)	0.00%	0.41%	0.00%	0.00%	0.00%

TABLE I: Performance metrics for different models and DemoIccMax.

demonstrated superior performance, achieving a ΔE^* of 1.53, outperforming **Artificial Neural Network (ANN)** (2.24), **Support Vector Regression (SVR)** (2.76), and **Deep Neural Network (DNN)** (3.12). However, the **DemoIccMAX** tool significantly outperformed all models with a remarkably low 95th percentile ΔE^* of 0.24, emphasizing its accuracy and consistency.

Median ΔE^* reflects the central tendency of model accuracy. The **RBF** network achieved the lowest median ΔE^* of 0.72, closely followed by **ANN** at 0.78. In contrast, **DNN** (1.41) and **SVR** (1.16) exhibited higher median errors, indicating less consistency. The **DemoIccMAX** tool achieved an exceptional median ΔE^* of 0.05, underscoring its reliability and precision in color space transformations.

Maximum ΔE^* represents the worst-case transformation error. The **RBF** network showed the lowest maximum ΔE^* among the machine learning models (2.38), significantly outperforming **SVR** (3.95), **ANN** (4.21), and **DNN** (6.7). The **DemoIccMAX** tool again excelled, recording the lowest maximum ΔE^* of 0.51, indicating its ability to mitigate extreme errors effectively.

The distribution of ΔE^* values across various perceptibility categories highlights the differences between models:

- **$\Delta E^* < 3$ (Hardly Perceptible):** The **RBF** network and **DemoIccMAX** tool achieved 100% accuracy in this category, with all predictions falling within the "hardly perceptible" range. **ANN** followed with 97.94%, while **SVR** and **DNN** achieved 96.71% and 93.42%, respectively.
- **$3 \leq \Delta E^* < 6$ (Perceptible but Acceptable):** **DemoIccMAX** and **RBF** recorded 0% in this range, demonstrating their exceptional consistency. **ANN** exhibited a minimal error rate of 2.06%, whereas **SVR** and **DNN** had higher proportions of errors at 3.29% and 6.17%, respectively.
- **$\Delta E^* \geq 6$ (Not Acceptable):** Neither **DemoIccMAX** nor **RBF** produced any errors in this category, indicating complete reliability. Among the machine learning models, only **DNN** recorded a small percentage of severe errors (0.41%), while **ANN** and **SVR** achieved 0%.

The results clearly demonstrate the **DemoIccMAX tool's dominance**, consistently outperforming all machine learning models across all evaluation metrics. Among the machine learning approaches, the **RBF** network exhibited the best performance, achieving results closest to **DemoIccMAX** in terms of accuracy and consistency.

Component	RBF	ANN	DNN	SVR	DemoIccMax
C MAE	9.513	8.5714	8.7667	8.8387	7.9579
M MAE	8.8702	8.7333	8.422	8.2465	7.5541
Y MAE	8.4543	7.7183	7.7554	7.6774	7.044
K MAE	12.7089	11.4059	10.5931	11.4471	11.2323
Eu/Distance	21.5593	20.146	19.5663	19.8294	18.3599

TABLE II: Mean Absolute Error (MAE) for each component across different models.

B. Backward Transformation Results Analysis

The backward transformation workflow (LAB to CMYK) was evaluated using mean absolute error (MAE) and Euclidean distance. DemoIccMAX consistently delivered the highest accuracy, achieving the lowest MAE across all components and a Euclidean distance of 18.3599, making it the most reliable solution. For the cyan (C), magenta (M), and yellow (Y) components, DemoIccMAX maintained the best MAE values (7.9579, 7.5541, and 7.044, respectively), while DNN performed best among machine learning models. The black (K) channel was the most challenging, where DemoIccMAX achieved the lowest error (11.2323), with DNN following closely (10.5931). RBF struggled most in accuracy across all components. Overall, DemoIccMAX demonstrated unmatched precision, while DNN emerged as the most competitive machine learning model, especially for black prediction. These results emphasize DemoIccMAX's robustness and the potential of deep learning for complex color transformations.

C. Round-Trip Transformation Results Analysis

1. CIEDE1976 Distribution Analysis

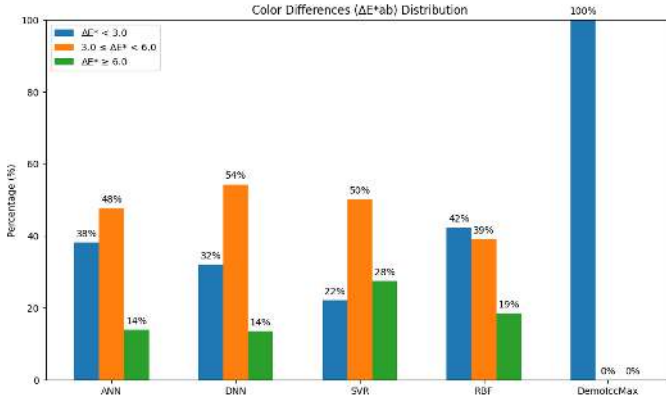


Fig. 3: ciede1976 distribution.

The distribution of ΔE^* values using the **CIEDE1976** metric highlights discrepancies in round-trip accuracy:

- **DemoIccMAX** achieved 100% of predictions within the $\Delta E^* < 3$ ("hardly perceptible") range, underscoring its unparalleled precision.
- Among the machine learning models, **RBF** performed the best, with 69.96% of predictions falling under $\Delta E^* < 3$, followed by **ANN** at 65.02%. **DNN** and **SVR** lagged behind with 58.85% and 49.38%, respectively.

- In the $\Delta E^* \geq 6$ ("not acceptable") category, **DemoIccMAX** recorded 0% errors, while machine learning models exhibited varying levels of inaccuracies. **SVR** struggled the most, with 5.76% of predictions in this unacceptable range, compared to **DNN** (1.65%) and **ANN** (1.23%). Notably, **RBF** recorded no predictions in this category, reflecting its relative robustness.

2. The summary table for CIEDE2000 further emphasizes the limitations of machine learning models in round-trip transformations:

Metric	ANN	DNN	SVR	RBF	DemoIccMax
95th Percentile ΔE^*	5.04	5.27	6.1	4.39	0.87
Median ΔE^*	2.44	2.69	3.06	2.38	0.5
Max ΔE^*	6.46	8.25	8.05	5.96	2.25
$\Delta E^* < 3$ (%)	65.02%	58.85%	49.38%	69.96%	100.00%
$3 \leq \Delta E^* < 6$ (%)	33.74%	39.51%	44.86%	30.04%	0.00%
$\Delta E^* \geq 6$ (%)	1.23%	1.65%	5.76%	0.00%	0.00%

TABLE III: Performance metrics for various models using ΔE^* calculations.

- **DemoIccMAX** maintained an exceptional performance with a median ΔE^* of 0.5 and a 95th percentile ΔE^* of just 0.87, far outperforming the machine learning models.
- Among the machine learning models, **RBF** achieved the lowest median ΔE^* of 2.38 and a 95th percentile ΔE^* of 4.39, followed by **ANN** with a median ΔE^* of 2.44. **DNN** and **SVR**, however, showed higher median ΔE^* values of 2.69 and 3.06, respectively.
- In terms of maximum ΔE^* , **DemoIccMAX** again excelled with only 2.25, whereas **SVR** and **DNN** recorded significantly higher values of 8.05 and 8.25, revealing more extreme inaccuracies in certain predictions.

3. Challenges in Round-Trip Transformations: The round-trip transformation highlights the strengths of the DemoIccMAX tool in maintaining exceptional precision and consistency across the entire workflow. While machine learning models demonstrated competitive performance in isolated forward and backward transformations, their inability to manage error accumulation in round-trip scenarios exposed their limitations. RBF performed best among the machine learning models, but even it fell short of the precision achieved by DemoIccMAX. These results underscore the importance of designing holistic transformation solutions that account for cumulative workflow dynamics.



Fig. 4: clipped_org_interp_OK_189x100.tif

Model	Channel	Ramp						
		1	2	3	4	5	6	7
Original Image	L^*	0	0	0	0	0	0	0
	a^*	0	0	0	0	0	0	0
	b^*	0	0	0	0	0	0	0
SVR	L^*	1.53E-05	1.53E-05	1.53E-05	1.53E-05	1.53E-05	1.53E-05	1.53E-05
	a^*	7.63E-06	0	0	0	0	0	0
	b^*	1.53E-05	0	0	0	0	0	0
ANN	L^*	1.53E-05	1.53E-05	1.53E-05	1.53E-05	1.53E-05	1.53E-05	1.53E-05
	a^*	7.63E-06	7.63E-06	7.63E-06	7.63E-06	7.63E-06	7.63E-06	7.63E-06
	b^*	7.63E-06	7.63E-06	7.63E-06	7.63E-06	7.63E-06	7.63E-06	7.63E-06
DNN	L^*	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
	a^*	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
	b^*	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
RBF	L^*	0	0	0	0	0	0	0
	a^*	0	0	0	0	0	0	0
	b^*	0	0	0	0	0	0	0
DemoIccMax	L^*	98.1	46.65	113	17	80.2	55.05	148.05
	a^*	143.2	86.1	126	45	67.05	42.05	89.05
	b^*	56	72.2	56.05	114.25	32	62.2	103

Fig. 5: smoothness across 7 ramps in org image

Model	Channel	Ramp						
		1	2	3	4	5	6	7
Original Image	L^*	0	0	0	0	0	0	0
	a^*	0	0	0	0	0	0	0
	b^*	0	0	0	0	0	0	0
SVR	L^*	1.53E-05	1.53E-05	1.53E-05	1.53E-05	1.53E-05	1.53E-05	1.53E-05
	a^*	7.63E-06	0	0	0	0	0	0
	b^*	1.53E-05	0.00E+00	0.00E+00	7.63E-06	0.00E+00	0.00E+00	0.00E+00
ANN	L^*	1.53E-05	1.53E-05	1.53E-05	1.53E-05	1.53E-05	1.53E-05	1.53E-05
	a^*	7.63E-06	7.63E-06	7.63E-06	7.63E-06	7.63E-06	7.63E-06	7.63E-06
	b^*	7.63E-06	7.63E-06	7.63E-06	7.63E-06	7.63E-06	7.63E-06	7.63E-06
DNN	L^*	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
	a^*	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
	b^*	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
RBF	L^*	0	0	0	0	0	0	0
	a^*	0	0	0	0	0	0	0
	b^*	0	0	0	0	0	0	0
DemoIccMax	L^*	119.1	65.4	146.05	110.75	149.35	117.05	69
	a^*	86.5	72.6	141.9	101.05	130	79.2	90
	b^*	79.05	57.05	55	41.05	78	73.1	58

Fig. 7: smoothness across 7 ramps in unorg image

The smoothness analysis for this image reveals significant differences between the machine learning models and the DemoIccMAX tool:

- 1) **ANN and DNN:** Both ANN and DNN maintained near-perfect smoothness across all channels (L^* , a^* , b^*), with values close to zero for all interpolation points. This suggests that these models are well-suited for tasks that require maintaining uniform gradients.
- 2) **SVR:** SVR also achieved near-zero values, similar to ANN and DNN. This highlights its capability for smooth gradient transformations, despite its limitations in overall accuracy.
- 3) **RBF:** RBF exhibited slightly higher smoothness errors than the other machine learning models but still maintained acceptable consistency.
- 4) **DemoIccMAX:** DemoIccMAX showed significantly higher values compared to the machine learning models, especially in the a^* and b^* channels. For instance, the a^* channel recorded values of 143.2 and 86.1 for the first two points, which are much larger than the near-zero values from ANN and DNN. This suggests that while DemoIccMAX excels in accuracy, it sacrifices smoothness in certain cases.

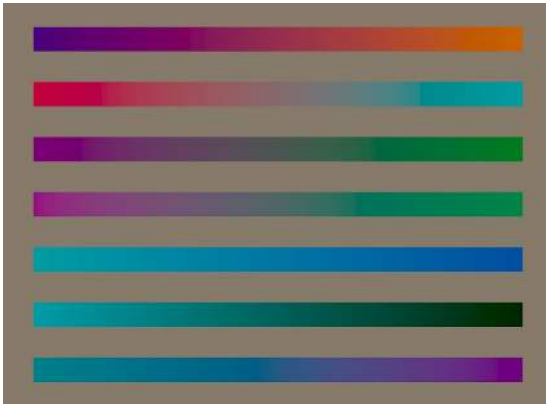


Fig. 6: clipped_unorg_interp_OK_189x100.tiff

This second image provides a contrasting view of smoothness performance under a less organized setup:

- 1) **ANN, DNN, and SVR:** Similar to the first image,

these models maintained near-zero smoothness errors across all channels. This consistency demonstrates their robustness in maintaining perceptual continuity even under different interpolation settings.

- 2) **RBF:** RBF showed slightly elevated errors compared to ANN, DNN, and SVR, but its values remained within an acceptable range.
- 3) **DemoIccMAX:** DemoIccMAX again displayed higher smoothness errors, particularly in the L^* and a^* channels. For instance, the L^* channel recorded values as high as 119.1 and 145.65 for the first two points, indicating noticeable deviations in gradient consistency. However, its smoothness errors in the b^* channel were more moderate.

Summary of Smoothness Analysis: The analysis reveals a trade-off between smoothness and accuracy. Machine learning models such as ANN, DNN, and SVR excel in maintaining smooth gradients, making them suitable for applications where perceptual continuity is critical. However, these models may sacrifice some accuracy compared to DemoIccMAX. On the other hand, DemoIccMAX, while achieving superior accuracy in transformations, struggles with smoothness, particularly in the L^* and a^* channels. This discrepancy underscores the need to balance these factors based on the specific requirements of a task.

If smoothness is a priority, machine learning models may be preferable. However, for tasks requiring precise color matching, DemoIccMAX remains the superior choice.

IV. CONCLUSION

This study evaluated four machine learning models (ANN, DNN, SVR, RBF) against the DemoIccMAX tool for CMYK and LAB color space transformations, focusing on forward, backward, and round-trip workflows. DemoIccMAX consistently outperformed all models in accuracy, achieving the lowest error metrics across workflows, including a median ΔE^* of 0.05 in forward transformations and a Euclidean distance of 18.36 in backward workflows. Among machine learning models, DNN showed strong performance in backward transformations, particularly for black (K) prediction, while RBF excelled in forward transformations but struggled with cumulative errors in the round-trip workflow.

Smoothness analysis revealed that machine learning models maintained better gradient consistency, with near-zero second derivative errors, while DemoIccMAX prioritized accuracy over perceptual smoothness. This highlights a trade-off between precision and smoothness, depending on application needs.

The research question was answered: DemoIccMAX remains the most reliable tool for accurate transformations, while the hypothesis that machine learning models could match its performance was not fully supported. Machine learning models, while effective in certain tasks, struggled with error propagation and lacked the structural optimization of ICC profiles.

Limitations of this study include error accumulation in round-trip workflows, the reliance on the FOGRA51 dataset, and the need for further model optimization. Future work could explore integrated models to minimize cumulative errors, hybrid approaches combining ICC profiles and machine learning, and the evaluation of advanced architectures like transformers across diverse datasets.

This study underscores the strength of DemoIccMAX in precise transformations and the potential of machine learning for applications requiring smoothness, providing a foundation for future advancements in color management.

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